

A Comparative Analysis of Synchronous USART and Asynchronous UART Communication Protocols

Authors: Meiting Dong (2022)

ABSTRACT

A rigorous comparison of synchronous and asynchronous serial communication protocols. The paper analyzes timing jitter, transmission latency, noise immunity, and CPU overhead in microcontrollers under different baud rates and payload sizes.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Justifies the choice of UART for inter-process communication in the EB15 robotic arm and guides the optimization of payload size.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Meiting Dong provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.